N. R. Sree Harsha

Theoretical and Computational Physicist | University of Rochester

PROFILE

- Theoretical and computational physicist specializing in non-equilibrium statistical mechanics, plasma physics, and electron-device modeling. Research develops Generalized Langevin Equation (GLE) and fluctuation-dissipation theorem (FDT) frameworks for kinetic-to-fluid coarse-graining, non-Markovian transport, ferroelectric switching, and hardware-aware stochastic optimization.
- Postdoctoral Associate, Department of Electrical & Computer Engineering, University of Rochester. Email: snaropan@ur.rochester.edu. Website: https://nrsreeharshae.github.io/harsha/
- Research record (Aug. 2026): 23 refereed journal articles, 1 book, 5 refereed conference papers; Co-Principal Investigator, NSF Award PHY-2606713.

EDUCATION

- PhD, Nuclear Engineering, Purdue University, 2018-2024. Thesis: Calculating Space-Charge-Limited Current Density in Nonplanar and Multidimensional Diodes. Advisor: Prof. Allen L. Garner.
- MSc, Nuclear Science and Engineering, University of Bristol, 2016-2018. Thesis: Application of Various Detection Techniques for Tritium Accountancy in DEMO.
- BE, Electrical and Electronics Engineering, R. V. College of Engineering, 2011-2015.

RESEARCH FUNDING

- National Science Foundation Award PHY-2606713, 2026-2029. Bridging Kinetic and Fluid Scales: A GLE-FDT Framework for Coarse-Graining Electron Kinetic Physics. Role: Co-Principal Investigator. PI: Prof. Chuang Ren; co-PI: Prof. Michael Huang; University of Rochester.

ACADEMIC AND RESEARCH EXPERIENCE

- Postdoctoral Associate, University of Rochester, Apr. 2025-present. Developing GLE-FDT plasma coarse-graining and EPOCH particle-in-cell validation; generalized Landau-Khalatnikov models for ferroelectric switching; and hardware-aware Langevin optimizers for NP-hard combinatorial optimization.
- Postdoctoral Research Assistant, Purdue University, June 2024-Mar. 2025. Applied non-equilibrium thermodynamics to collisional space-charge-limited current; extended the Mott-Gurney law to multidimensional geometries; benchmarked against MICHELLE and VSim; collaborated on non-isometric thermal deformation of composite laminates.
- Graduate Research Assistant, Purdue University, Jan. 2021-Apr. 2024. Derived exact and approximate SCLC solutions for nonplanar and multidimensional diodes using conformal mapping, variational calculus, and Lie-point symmetries.
- Graduate Research Assistant, University of Bristol and Culham Centre for Fusion Energy, Sept. 2016-Jan. 2018. Developed a beta-induced X-ray spectrometry technique for tritium accountancy in the DEMO fusion power plant using Geant4.
- LIDAR for Advanced Gas-Cooled Reactors, EDF Energy, Sept. 2016-Sept. 2017. Designed a LIDAR-based imaging and scanning system for graphite-brick defect detection.

HONORS AND RECOGNITION

- IOP Trusted Reviewer status, IOP Publishing (2024).
- Purdue Nuclear Graduate Student Research Recognition Seminar Award (2024).

HONORS AND RECOGNITION (continued)

- Purdue Nuclear Engineering Graduate Research Award (2023).

INVITED SEMINARS

- N. R. S. Harsha, Calculating Space-Charge Limited Current Density in Nonplanar and Multi-dimensional Diodes, Los Alamos National Laboratory, 21 June 2023.
- N. R. S. Harsha, Application of Symmetries for Calculating Space-Charge-Limited Current, Graduate Research Recognition Seminar, Purdue University, 17 Jan. 2024.

REFEREED JOURNAL ARTICLES

1. A. L. Garner, J. K. Wright, and N. R. S. Harsha, A capacitance-based approach for determining the space-charge limited current density and electric potential in a crossed-field gap, AIP Advances 16, 065213 (2026).
2. N. R. S. Harsha, X. Zhu, D. Smithe, M. E. McFeely, A. Panda, and A. L. Garner, A general formulation of multi-dimensional space-charge-limited current density with nonzero monoenergetic initial velocity, Journal of Physics D: Applied Physics 59, 215502 (2026).
3. J. K. Wright, N. R. S. Harsha, and A. L. Garner, A generalized equation for the critical current for a one-dimensional crossed-field gap in an orthogonal coordinate system, Physics of Plasmas 32, 072115 (2025).
4. A. L. Garner, N. R. S. Harsha, and A. M. Loveless, Mott-Gurney for nonplanar and multi-dimensional diodes using point transformations, Journal of Applied Physics 137, 234501 (2025).
5. A. L. Garner and N. R. S. Harsha, The implications of collisions on the spatial profile of electric potential and the space-charge limited current, IEEE Transactions on Electron Devices 72, 1419-1426 (2025).
6. J. M. Halpern, N. R. S. Harsha, A. M. Darr, and A. L. Garner, Space-charge-limited current for nonplanar relativistic diodes with nonzero monoenergetic initial velocity using point transformations, Physics of Plasmas 32, 012106 (2025).
7. A. G. Sinelli, L. I. Breen, N. R. S. Harsha, A. M. Darr, A. M. Komrska, and A. L. Garner, Theoretical assessment of the transition between electron emission mechanisms for nonplanar diodes, IEEE Transactions on Electron Devices 72, 397-403 (2025).
8. N. R. S. Harsha, K. Chinwicharnam, V. Doty, and R. Byron Pipes, Non-isometric deformation in double curvature and symmetric composite laminates due to shrinkage, Polymer Composites 45, 16589-16594 (2024).
9. X. Zhu, J. K. Wright, N. R. S. Harsha, and A. L. Garner, Critical current in a two-dimensional non-magnetically insulated crossed-field gap with monoenergetic emission, Physics of Plasmas 31, 072101 (2024).
10. X. Zhu, J. K. Wright, N. R. S. Harsha, J. Browning, and A. L. Garner, Electron trajectories and critical current in a two-dimensional planar magnetically insulated crossed-field gap, IEEE Access 12, 11378-11387 (2024).
11. X. Zhu, N. R. S. Harsha, and A. L. Garner, Uniform space-charge-limited current for a two-dimensional planar emitter with nonzero monoenergetic initial velocity, Journal of Applied Physics 134, 113301 (2023); Erratum 134, 239902 (2023).

REFEREED JOURNAL ARTICLES (continued)

12. N. R. S. Harsha, J. M. Halpern, A. M. Darr, and A. L. Garner, Space-charge-limited current density for nonplanar diodes with monoenergetic emission using Lie-point symmetries, Physical Review E 106, L063201 (2022).
13. J. M. Halpern, A. M. Darr, N. R. S. Harsha, and A. L. Garner, A coordinate system invariant formulation for space-charge limited current with nonzero injection velocity, Plasma Sources Science and Technology 31, 095002 (2022).
14. A. L. Garner, A. M. Darr, and N. R. S. Harsha, A tutorial on calculating space-charge limited current density for general geometries and multiple dimensions, IEEE Transactions on Plasma Science 50, 2528-2540 (2022).
15. N. R. S. Harsha, M. Pearlman, J. Browning, and A. L. Garner, A multi-dimensional Child-Langmuir law for any diode geometry, Physics of Plasmas 28, 122103 (2021).
16. A. M. Darr, N. R. S. Harsha, and A. L. Garner, Response to Comment on A coordinate system invariant formulation for space-charge limited current in vacuum, Applied Physics Letters 119, 206102 (2021).
17. N. R. S. Harsha and A. L. Garner, Analytic solutions for space-charge-limited current density from a sharp tip, IEEE Transactions on Electron Devices 68, 6525-6531 (2021).
18. A. M. Darr, N. R. S. Harsha, and A. L. Garner, Response to Comment on A coordinate system invariant formulation for space-charge limited current in vacuum, Applied Physics Letters 118, 266102 (2021).
19. N. R. S. Harsha and A. L. Garner, Applying conformal mapping to derive analytical solutions of space-charge-limited current density for various geometries, IEEE Transactions on Electron Devices 68, 264-270 (2021).
20. N. R. S. Harsha, A review of the variational methods for solving DC circuits, European Journal of Physics 40, 033001 (2019).
21. N. R. S. Harsha, The tightly bound nuclei in the liquid drop model, European Journal of Physics 39, 035802 (2018).
22. N. R. S. Harsha, A singularity in Kirchhoff’s circuit equations, European Journal of Physics 37, 055801 (2016).
23. N. R. S. Harsha, A. Sreedevi, and A. Prakash, An unsolved electric circuit: a common misconception, Physics Education 50, 568 (2015).

BOOK

- N. R. S. Harsha, A. Prakash, and D. P. Kothari, The Foundations of Electric Circuit Theory, IOP Publishing, Bristol, UK (2016).

PREPRINT AND CURRENT MANUSCRIPTS

- N. R. S. Harsha, Z. Yu, C. Ren, V. Billings, and M. Huang, Nonlinear dynamical friction from the Doppler-shifted equilibrium memory kernel, arXiv:2602.04545 (2026).
- N. R. S. Harsha and M. Huang, A Generalized Landau-Khalatnikov Model for Ferroelectric Switching, manuscript (2026).
- N. R. S. Harsha, Non-Markovian Space-Charge-Limited Transport: Steady-State Scaling and Device Admittance, manuscript (2026).

PREPRINT AND CURRENT MANUSCRIPTS (continued)

- N. R. S. Harsha, Noise-Fidelity-Constrained Passive Approximation of Constant-Phase Elements, manuscript (2026).

SELECTED TEACHING AND SERVICE

- Instructor of Record, MA 158 (Precalculus), Department of Mathematics, Purdue University, Fall 2020. Sole instructor for 40-50 first-year undergraduates.
- Teaching Assistant, MA 163 (Calculus I), Purdue University, Fall 2019.
- Invited Guest Lecturer, NUCL 200, Purdue University, Jan. 2023.
- Referee for 15 international journals, including Classical and Quantum Gravity, IEEE Transactions on Plasma Science, Journal of Applied Physics, Journal of Physics D: Applied Physics, Physics of Plasmas, and Plasma Sources Science and Technology.
- Professional memberships: Institute of Electrical and Electronics Engineers (IEEE); American Physical Society (APS).